Olist E-Commerce Analytics — Capstone Report

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1. Project Objective

Build an end-to-end analytics pipeline for the Olist Brazilian e-commerce dataset using Databricks Bronze, Silver and Gold layers, then define the required Gold-to-Snowflake handover for analytical SQL.

2. Business Questions

• Which product categories have the worst delivery performance and how does it relate to review scores?

• What share of revenue comes from repeat customers and how does it change quarter by quarter?

• Which sellers have high late-delivery rates and how does seller lateness relate to review scores?

3. Pipeline

Bronze loads nine Olist CSV datasets into Delta tables. Silver standardizes timestamps, handles duplicates, aggregates review/payment information, joins customer/seller/product/category information, and derives delivery, lateness, quarter and revenue fields. Gold creates three business tables for category delivery, customer revenue and seller reliability.

4. Data Profile

Dataset	Rows
Orders	99,441
Customers	99,441
Order items	112,650
Payments	103,886
Reviews	104,162
Products	32,951
Sellers	3,095
Geolocation	1,000,163
Category translation	71
Silver analytical rows	113,425

5. Findings

Q1: In the displayed Gold output, home_comfort_2 has a 16.67% late rate, furniture_mattress_and_upholstery 13.16%, and audio 12.64%. office_furniture has a displayed average delivery time of 20.79 days and average review score of 3.49.

Q2: The customer-level Gold output tracks repeat-customer share by customer count, while the SQL analysis calculates repeat-customer revenue share. Revenue share is 3.48% in 2016-Q4, 7.35% in 2017-Q2, 7.34% in 2017-Q3 and 4.62% in 2018-Q3. 2018-Q4 has zero revenue and is not meaningful for revenue-share interpretation.

Q3: The displayed seller results include a seller with a 50.00% late rate and 1.72 average review score, and another with a 48.94% late rate and 2.57 average review score. Sellers are filtered to at least 10 orders.

6. Job Evidence

The Databricks Job uses three dependent Serverless tasks: Bronze → Silver → Gold. The supplied run evidence shows all three tasks succeeded.

7. Snowflake Status

The capstone brief requires a Gold-to-Snowflake stage/COPY INTO handover and COPY_HISTORY evidence. A working Snowflake account was not available during implementation. The repository therefore contains an explicit handover template but does not claim a completed Snowflake load.

8. Repository Contents

The repository contains exported notebooks, SQL analysis files, a Snowflake handover template, architecture diagram, job evidence, hosted project page and demo script.